

Lineage-dependent perturbation of aging-associated methylation patterns in Fanconi anaemia

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Background

- For risk of cardiovascular disease, blood pressure can be used as a readout in a trial of an intervention.
- The DEB test is the current gold-standard assay for chromosomal breakage.
- It is now well established that Fanconi individuals show a premature aging phenotype.^[1]
- DNA cytosine methylation (DNAm) is also now well established as an accurate readout of biological age.
- We hypothesise that biological age of minimally-invasive samples from FA individuals could provide a readout for trials of hypothetical interventions, following recent related work.^[2]

Overview of study

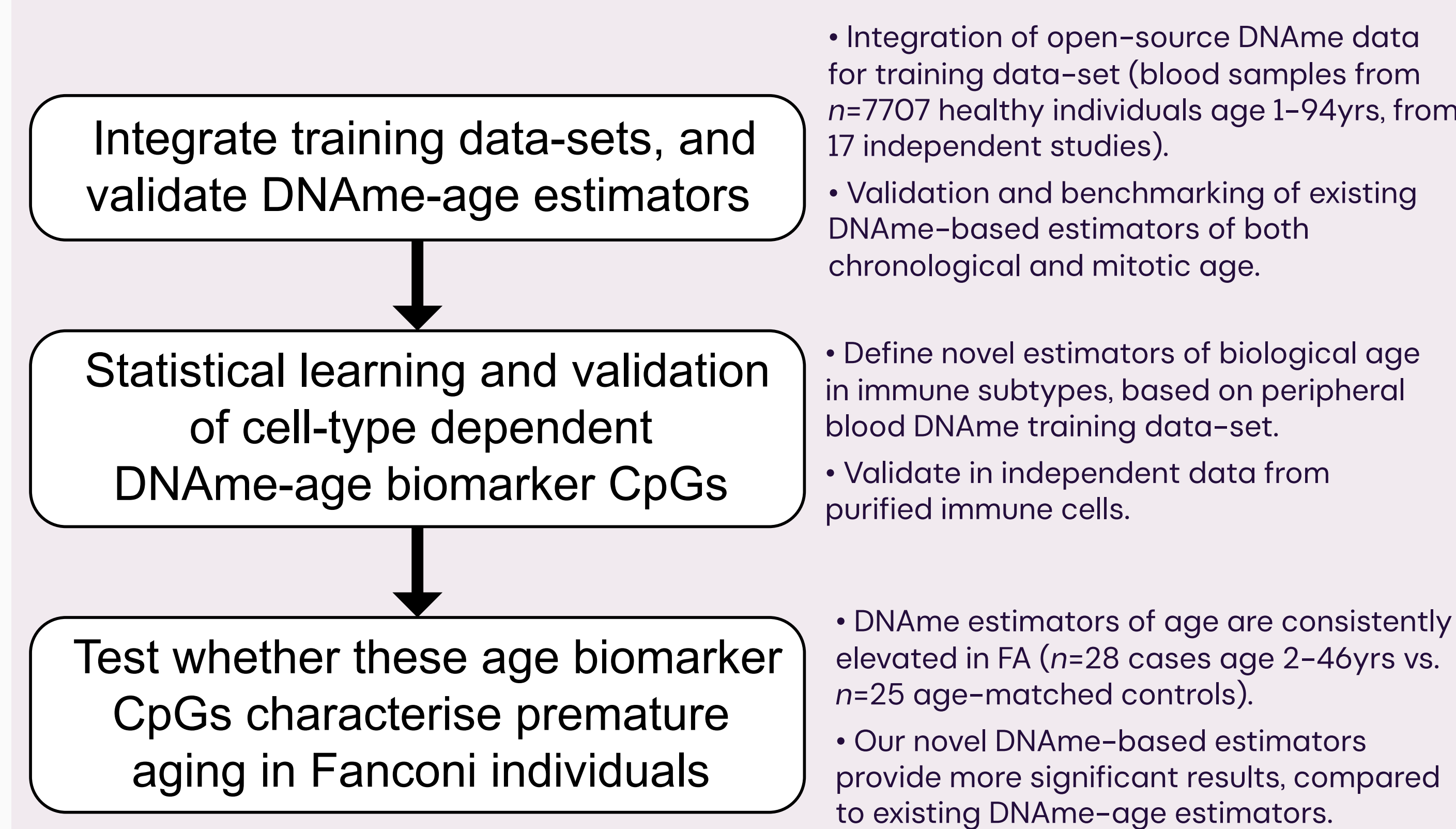


Figure 1: Overview of study.

Estimating chronological age with DNAm clocks

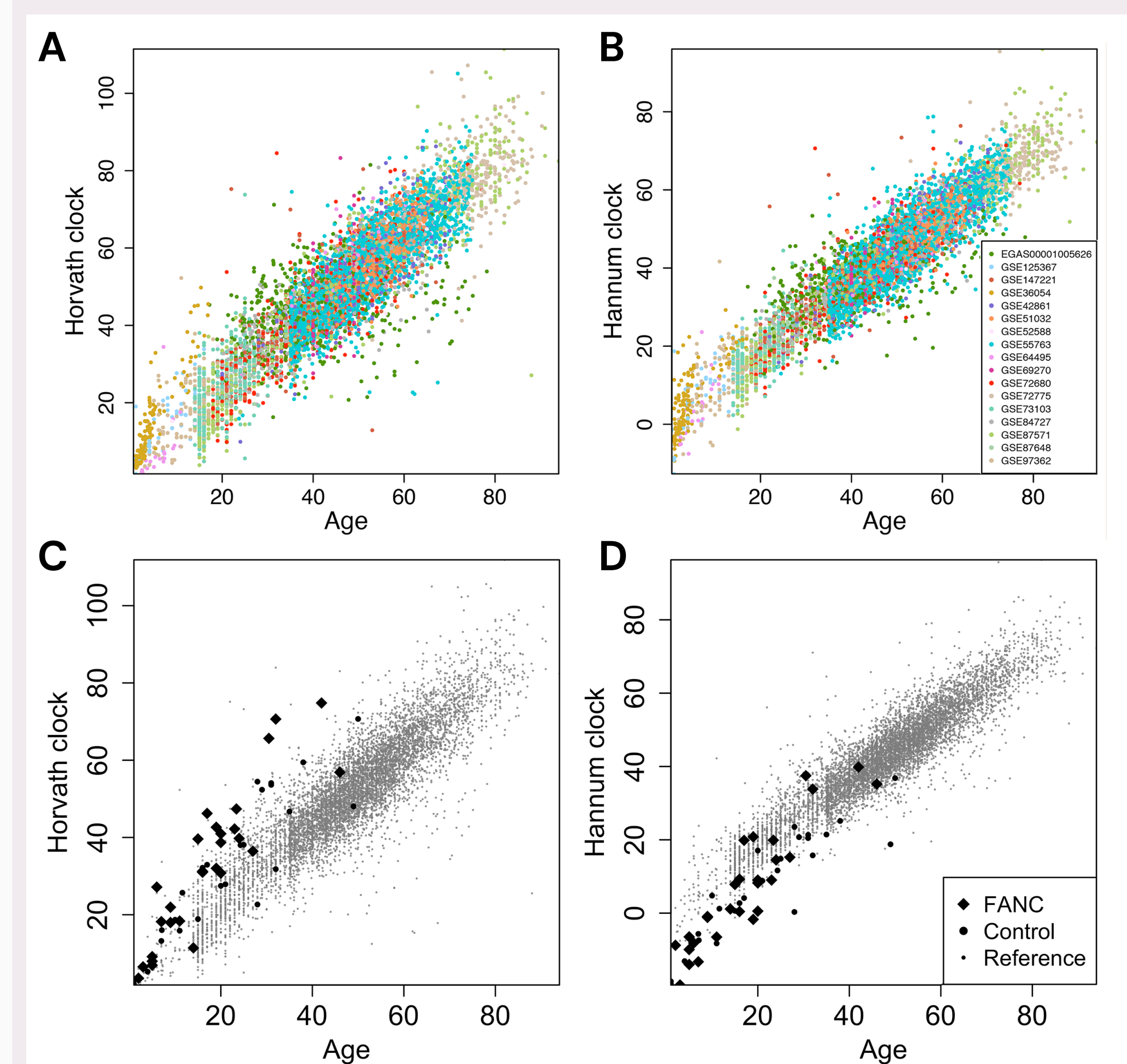


Figure 2: The original DNAm-age estimators, the (a) Horvath^[3] and (b) Hannum clocks^[4], show excellent correspondence with chronological age across the 17 integrated data-sets, $n=7707$. The Horvath and Hannum clocks are significantly elevated in FA individuals ($n=28$) compared to controls ($n=25$) after adjusting for chronological age: (c) Horvath $p=0.013$ and (d) Hannum $p=0.055$.

Mitotic age aims to estimate the average number of cell divisions that preceded the observed cells in the sample

- 2006 PCGTs repressed by histone modification:** Lee et al characterised the role of polycomb group target genes (PCGTs) in embryonic development, especially via H3K27me3 repression of developmental lineage markers.^[5]
- 2008 PCGTs repressed in cancer:** Ben-Porath et al showed that PCGTs are repressed in cancer, contributing to the return of stem-like cell characteristics.^[6]
- 2010 PCGT hypermethylation with age:** Teschendorff et al showed that the promoters of PCGTs are progressively hypermethylated with age.^[7]
- 2016 DNAm mitotic age estimator:** Lee et al used PCGT hypermethylation to estimate the cumulative number of cell divisions that preceded the observed cells (averaged over the sample), referred to as the 'mitotic age'.^[8]

DNAm-age is elevated in FA – comparison of results

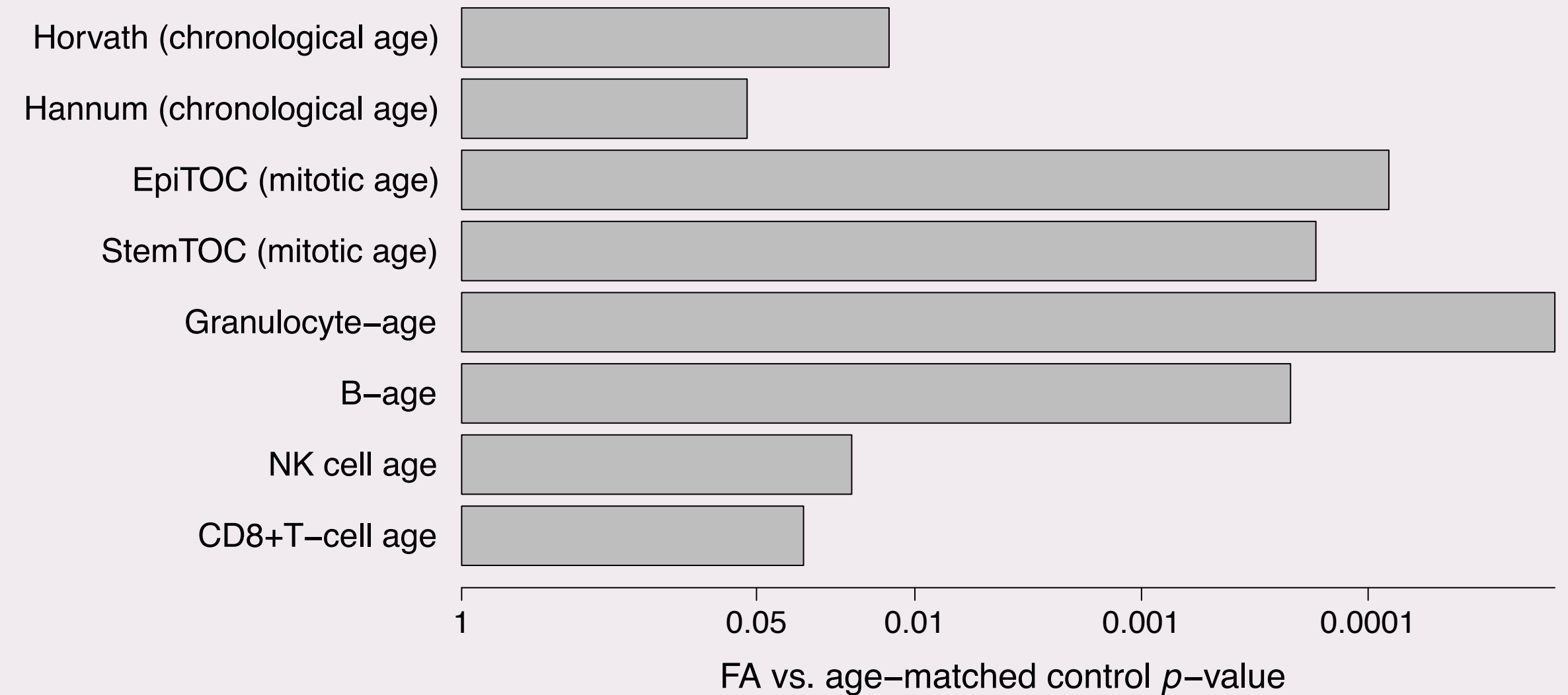


Figure 3: Significances of increase in DNAm-age in FA individuals compared to controls.

- The sizes of the bars show significance levels of increases in DNAm-age estimators in FA individuals compared to age-matched controls.

- Our novel granulocyte-age biomarker shows the greatest level of statistical significance.

DNAm-age estimators: gene-set enrichment analysis

A		Gene set	OR	FDR-p
BENPORATH_ES_WITH_H3K27ME3	7.6	5.0E-7		
BENPORATH_EED_TARGETS	7.8	5.0E-7		
BENPORATH_SUZ12_TARGETS	7.7	5.0E-7		
MIKKELSEN_MCF_HCP_WITH_H3K27ME3	9.6	2.3E-6		
MEISSNER_NPC_HCP_WITH_H3K4ME2_AND_H3K27ME3	12.7	2.5E-6		
MIER1_TARGET_GENES	12.7	8.9E-6		
BENPORATH_PRC2_TARGETS	8.0	3.8E-5		

B		Gene set	OR	FDR-p
BENPORATH_ES_WITH_H3K27ME3	3.4	1.8E-47		
MIKKELSEN_MCF_HCP_WITH_H3K27ME3	4.0	1.3E-35		
BENPORATH_SUZ12_TARGETS	3.0	3.4E-34		
BENPORATH_EED_TARGETS	3.0	1.9E-33		
MIKKELSEN_MCF_HCP_WITH_H3K27ME3	4.3	4.7E-31		
BENPORATH_PRC2_TARGETS	3.3	1.1E-26		
GOBP_NEUROGENESIS	2.2	1.9E-25		
GOBP_GENERATION_OF_NEURONS	2.3	2.6E-24		
MEISSNER_BRAIN_HCP_WITH_H3K4ME3_AND_H3K27ME3	2.9	5.0E-24		

Table 1: Gene-set enrichment analyses for 35361 Broad Institute MSigDB gene-sets^[9] (a) For granulocyte-age biomarker CpGs, and (b) for CpGs hypermethylated in FA individuals (compared to controls) after adjusting for chronological age.

- The Ben-Porath gene-sets are the same PCGT gene-sets that the original mitotic age estimators are based on. Out of 35361 gene-sets, the Ben-Porath gene-sets are the most significantly enriched gene-sets both for CpGs hypermethylated in FA (adjusted for age), and for our cell-type dependent age-biomarker CpGs.
- This suggests that the most prominent FA-age and cell-type age DNAm alterations result from hypermethylation of PCGTs, and more specifically hypermethylation of bivalent genes.

DNAm-age estimators quantify accumulated chromatin stress?

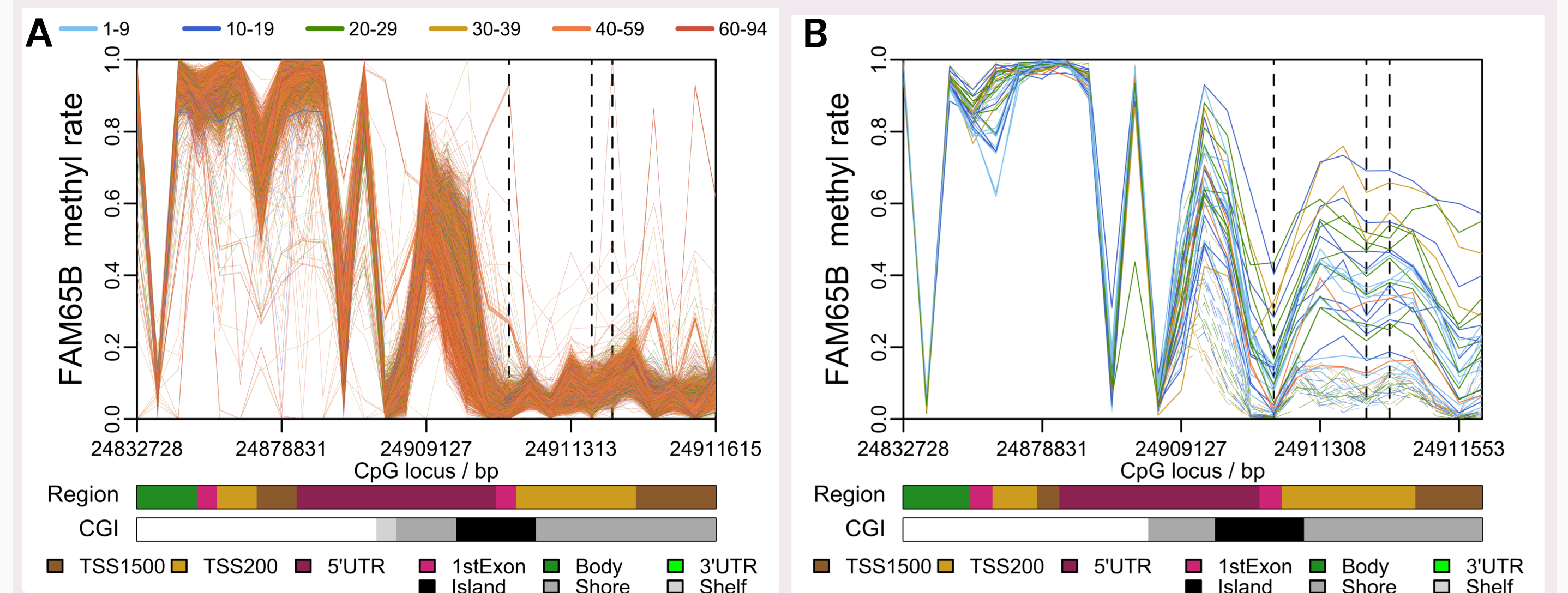


Figure 4: DNAm profiles for (a) Training data-set ($n=7707$) and (b) FA individuals ($n=28$) and age-matched controls ($n=25$) for an example target-gene of interest; colours of lines indicate ages of individuals.

- We found granulocyte-age, CD8T-age, and NK-age promoter hypermethylation for FAM65B, as well as promoter hypermethylation for this gene in FA individuals. FAM65B is a FOXO1 target; its function is to inhibit the RhoA enzyme, which has a role in DSB repair^[10]. Like many PCGTs, FAM65B is a bivalent gene^[11], and it was recently independently reported as being functionally downregulated via hypermethylation in Fanconi individuals.^[12]

- Key observation: The most prominent genome-wide CpG hypermethylation signal both in FA vs. control and in our cell-type dependent age-biomarkers, corresponds to CpGs located in the promoters of PCGTs, and more specifically of bivalent genes.

- The expression of bivalent genes is controlled by the balance of H3K27me3 (repressing) vs. H3K4me3 (activating) chromatin modifications.

- How can we explain this observation? Does repression via DNAm replace repression via H3K27me3, if histone modification becomes disrupted due to chromosomal instability?

References

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